Highway Engineering

- INS3441 -

ASSIGNMENT



Subject

The calculations and drawings will be prepared for the highway being designed between given A and C points on a contour map with a scale of 1/2.000. The first necessary point will be named as “A”, while the second one as “C”. The direction of the designed highway is going to be from point “A” to point “C”.

A km $0+000^{00}$

C km $0+\ldots\ldots^{\cdots}$

Given

- Digital contour map (Scale: 1/2.000)
- Project criteria

Project velocity; Design speed (V_p)	= 70 km/h
Platform width (B)	= 10 m (2 x 3.5 +2 x 1.5)
Maximum longitudinal grade (g_{max})	= 8%
Right of way width	= 40 m
Minimum horizontal curve length (L)	= 50 m
Maximum superelevation (q_{max})	= 8%
Maximum widening of the horizontal curve (b_w)	= 2.0 m
Minimum vertical curve radius	= R_{CREST} = 4000 m = R_{SAG} = 2000 m

Before Starting

- Download documents about assignment → https://avesis.yildiz.edu.tr/yardim/dokumanlar password: **INS3441_2122OD**
- Download the file named as «2021-2022 Karayolu Müh. Ödev Pafta Numaraları Dağıtımı»

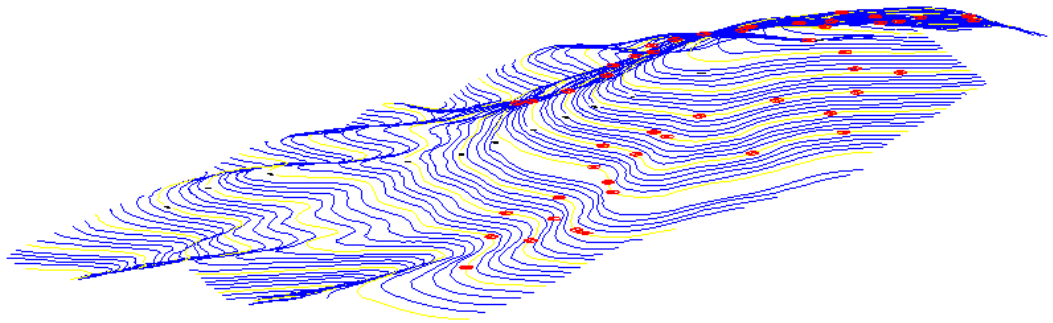
→Check your sheet number and starting (A) and ending (C) points of your road.

→Download your digital contour map (autocad file).

- Download the file named as «Karayolu Müh. Ödevinde Kullanılacak Tanjant Final-Tanjant Orijin Fontu». In order to show the symbols ⱷ and Ⱶ install this font to your computer. Use the font named as «NewSymbolFont» and type 1 for ⱷ; 2 for Ⱶ.
- Also some documents for English course may be downloaded → https://avesis.yildiz.edu.tr/ozen/dokumanlar password: **INS3441_1**

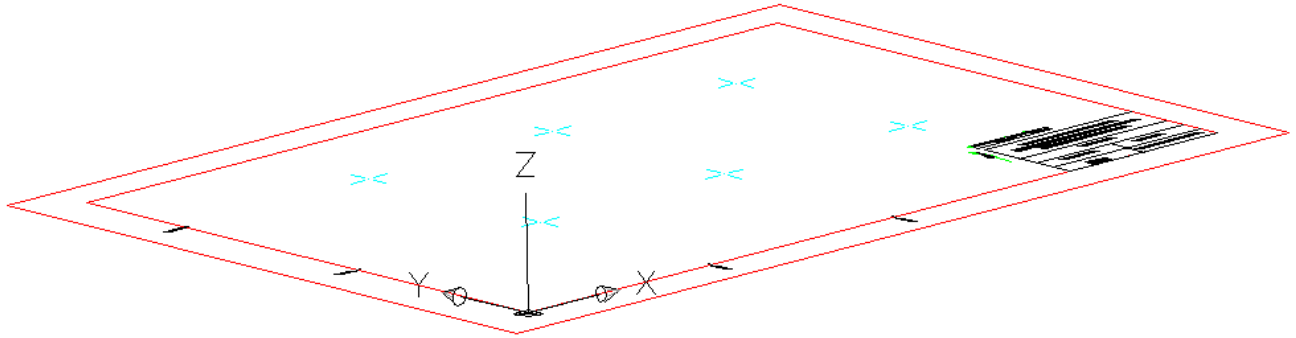
Preparing of the Digital Contour Map

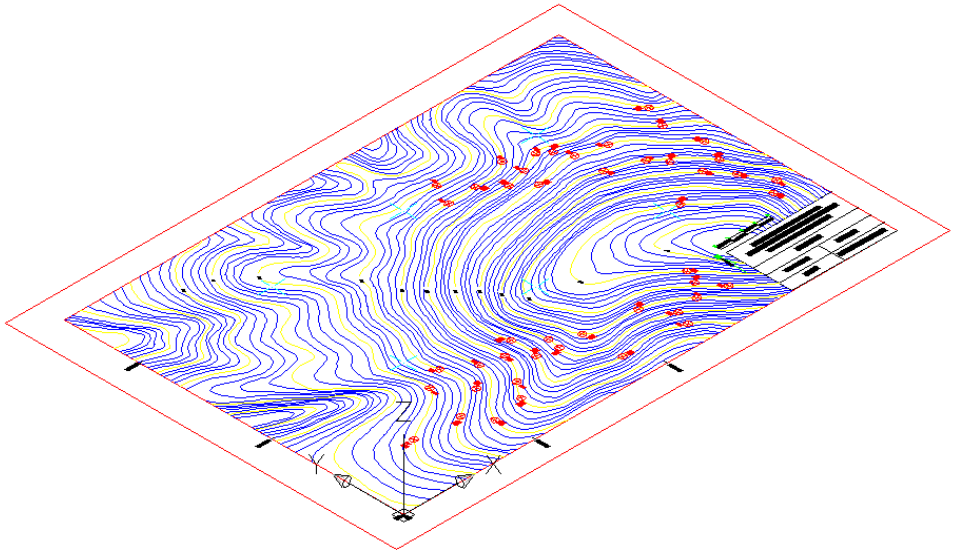
- Before starting your project, you should be sure that Z coordinates of all elements in the map are zero.



Choose All → Properties:

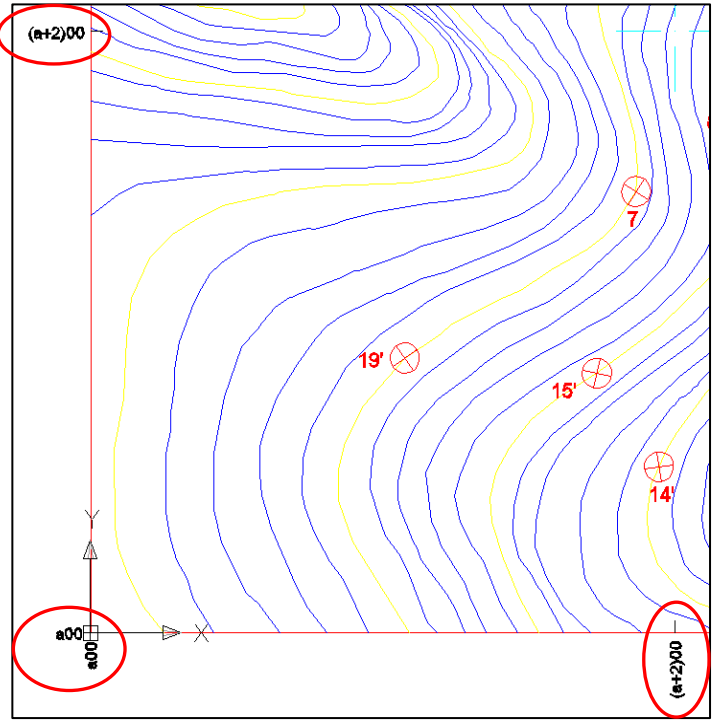
- *Line* → *Start Z=0* and *End Z=0*
- *Polyline* → *Elevation=0*
- Others → *Position Z=0*





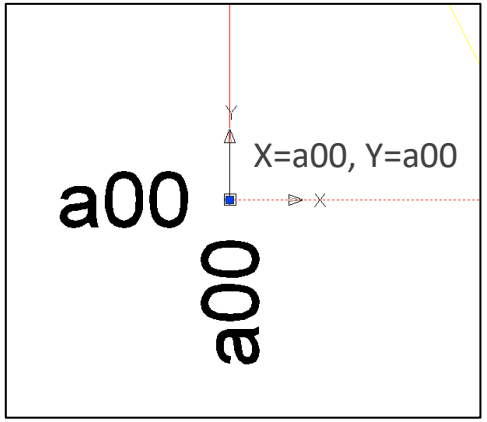
Preparing of the Digital Contour Map

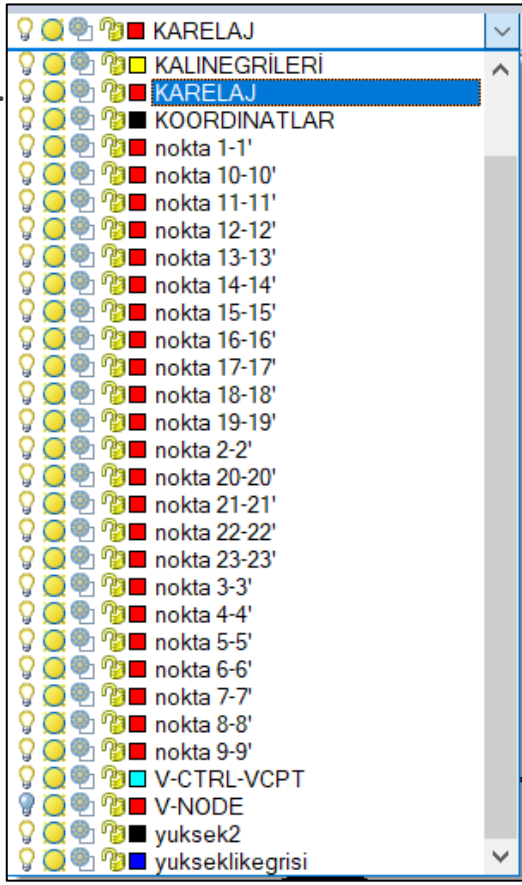
- The given autocad files are scaled as 1/1 no need to change scale. You will use 1 unit-length in autocad for 1 meter-length in real.
- Edit the coordinates of your map based on your student ID. «a» is the last digit of your student ID.
- Edit the title block of your map; Name-Surname, SheetName (Temporary Plan, Temporary Profile etc.), so on.



- Move your map by moving corner point to the coordinate «a00,a00»
- Turn off unnecessary layers.

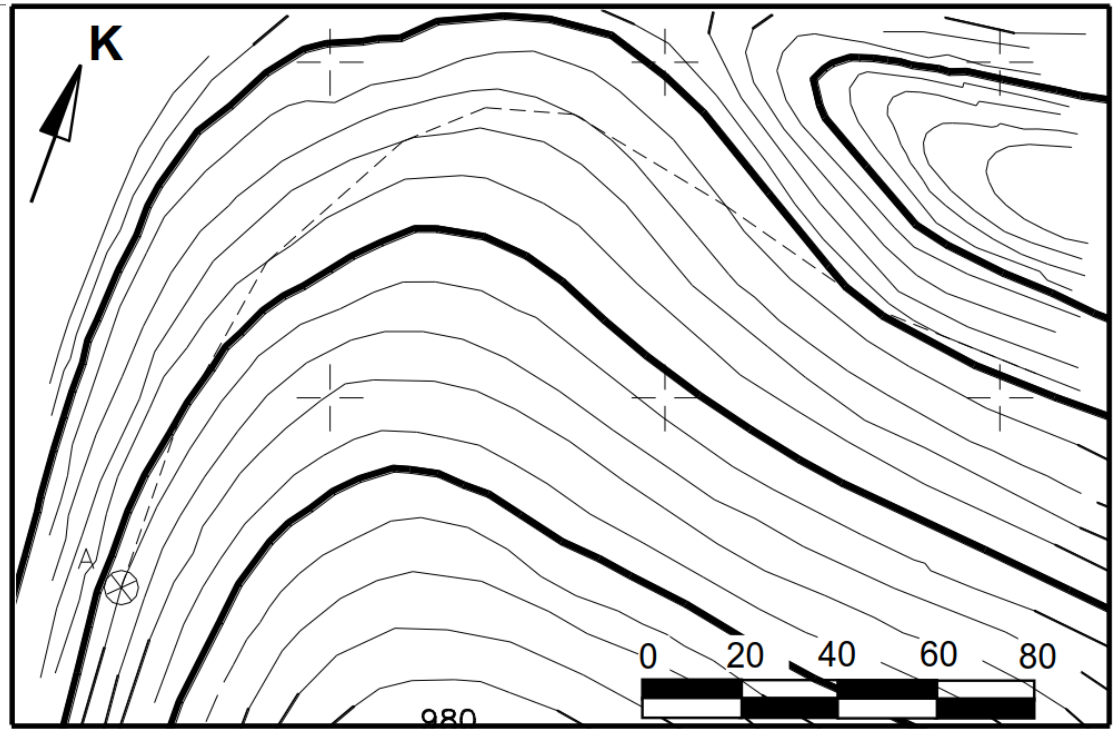
YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PAFTA ADI	
AD-SOYAD	NUMARA
GRUP	PAFTA NO:1-1 / 2000





Zero Line Study

- The polygon which makes the earthwork zero on the axis will be determined by the zero line study followed between the given points.
- The polygon lines should not intersect or be tangent with the contour lines.
- Zero line will be drawn as a dashed line and with a lineweight of 0.3 mm.



- The aim of the zero line study is to go from point A to point C by a constant slope (S_0). Therefore the contour lines should be passed by a lines with equal length.

$$S_0 = \frac{\Delta H(A-C)}{L(A-C)} \times 100 \rightarrow for\ example: \frac{860-840}{600} \times 100 = \%3.33 < \%8 \quad proper!$$

($L_{A\text{-}C}$ is taken as an approximate value)

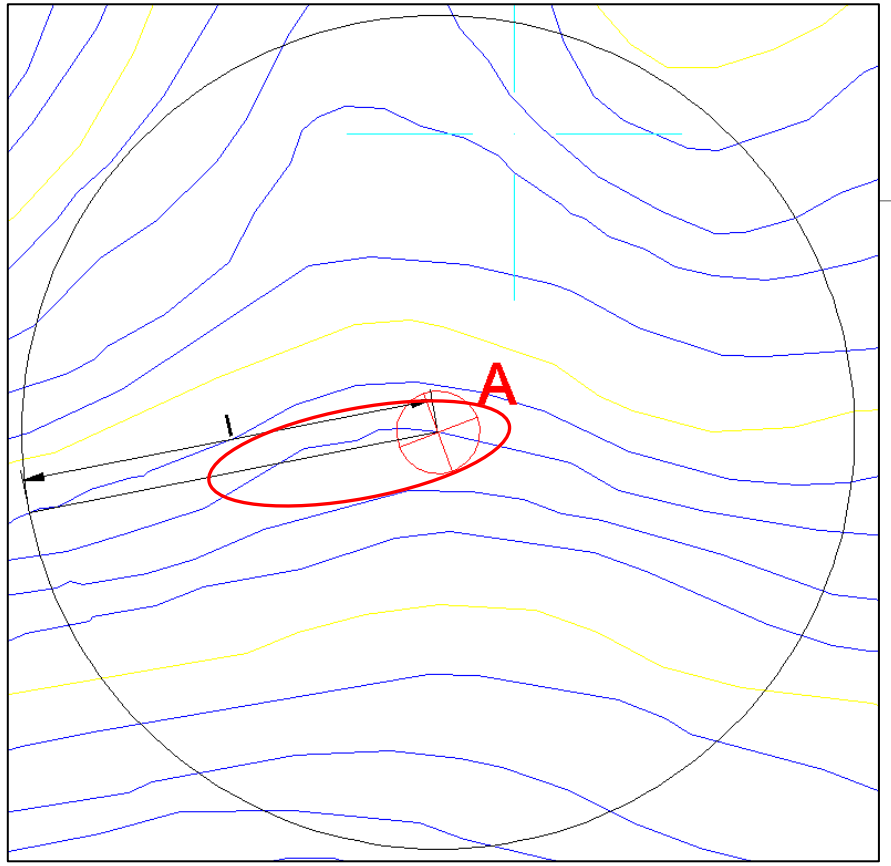
(The elevation difference between A-C (ΔH) have to be 20 meter)

- Use the formula below to find the length of the lines;

$$l_0 = \frac{h}{S_0} \times 100 \rightarrow for\ example: \frac{2}{3.33} \times 100 = 60.60\ m$$

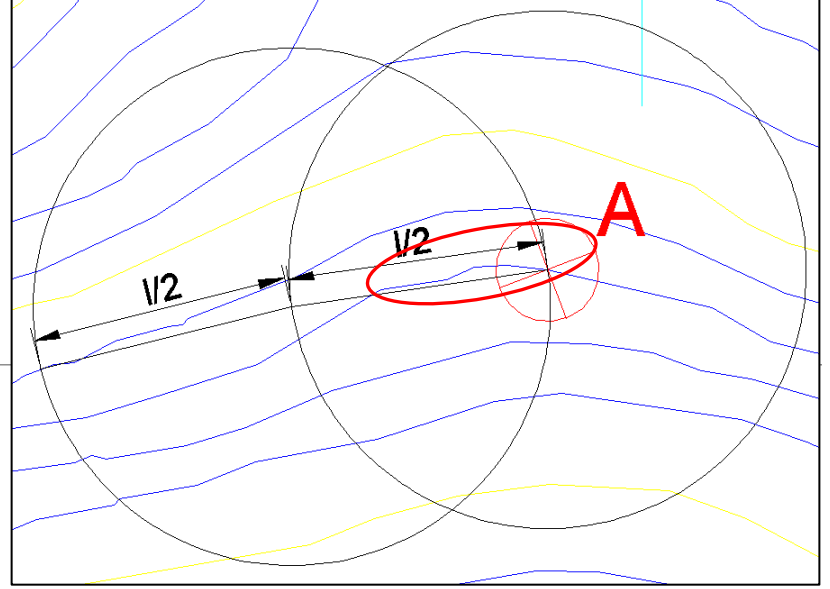
l_0 is the radius of the circle on the autocad drawings scaled as 1/1. For printout, you have to find compasses span value.

$$On\ map \rightarrow l_p = l_0 \times \frac{1}{2000} \rightarrow for\ example: 60.60 \times \frac{1}{2000} = 0.03\ m = 3\ cm$$



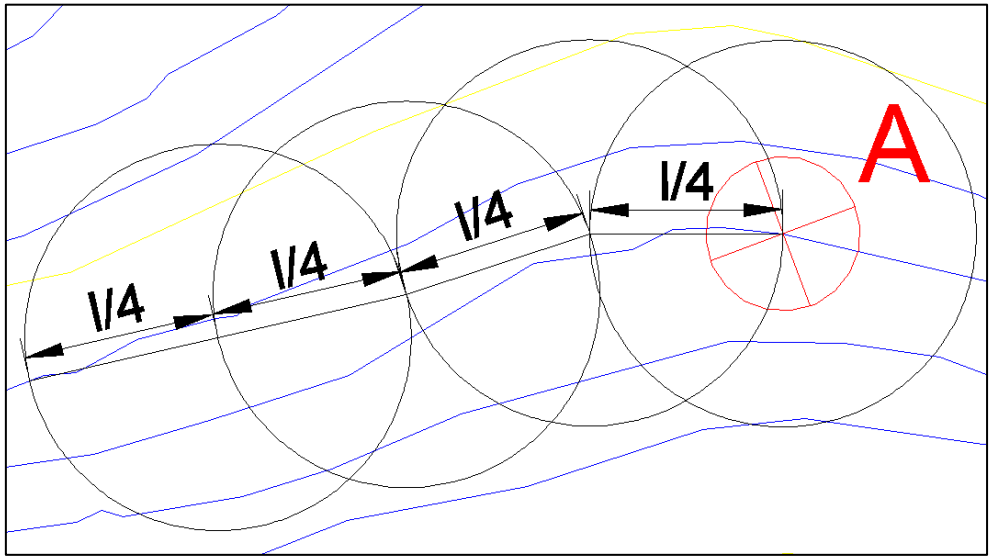
98
96

s (%) = (h/l) × 100



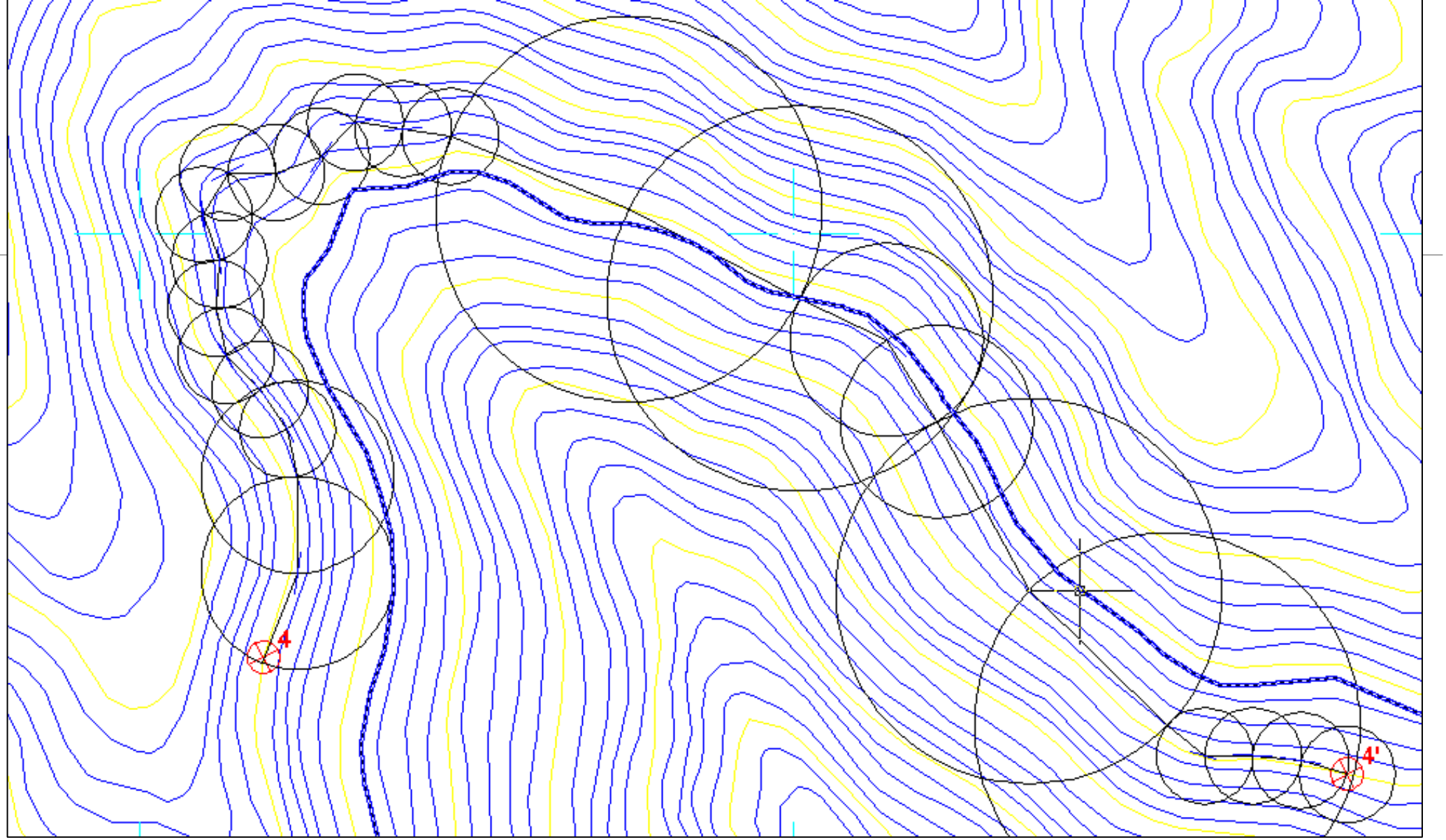
98
97
96

s (%) = [(h/2)/(l/2)] × 100



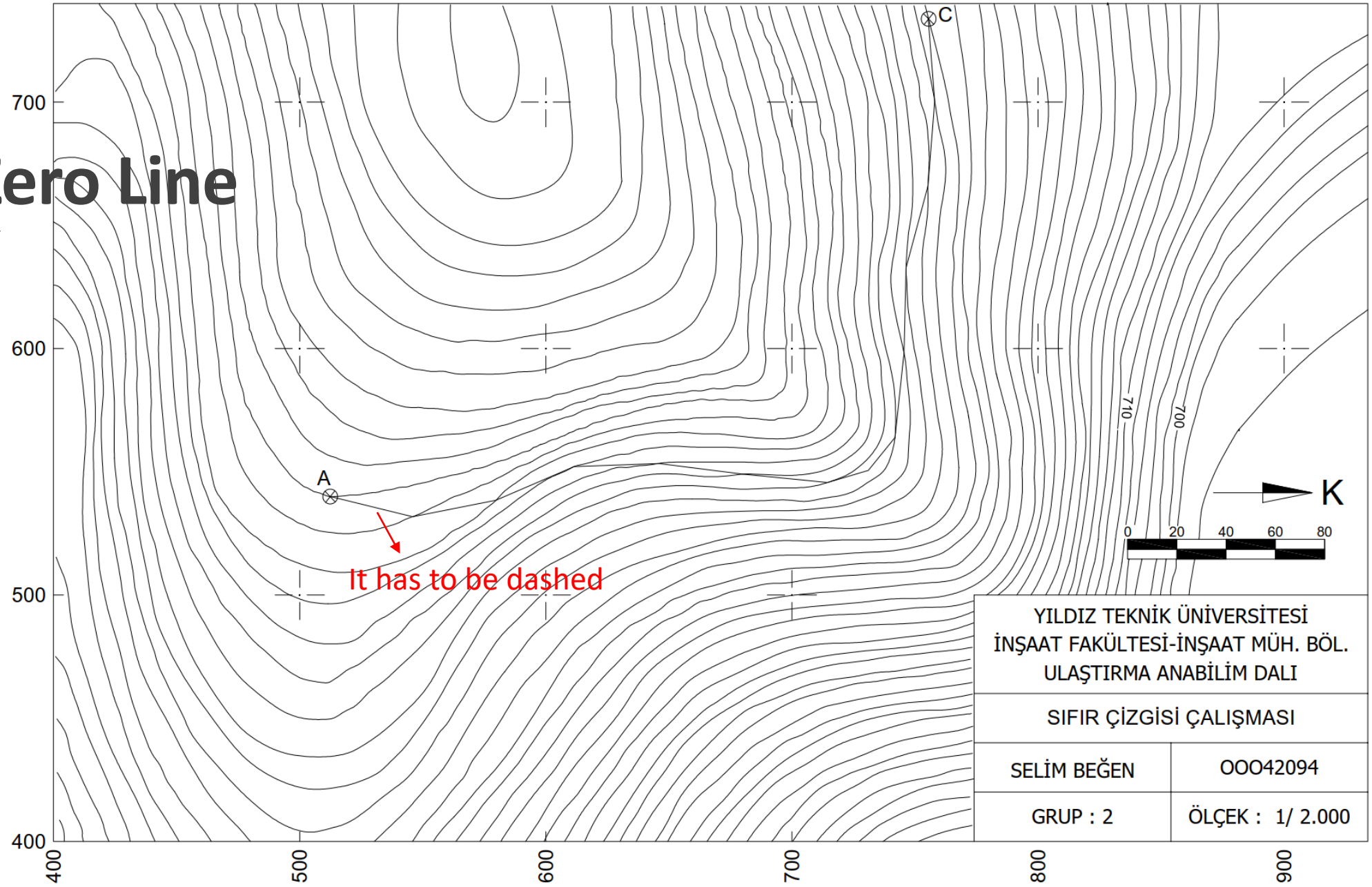
98
97.5
97
96.5
96

s (%) = [(h/4)/(l/4)] × 100



Zero line should go to the contour line above or below. You may divide the line into two pieces if it intersects with contour line. If necessary you may divide into 4 pieces but keep the total l_0 constant.

Completed Zero Line

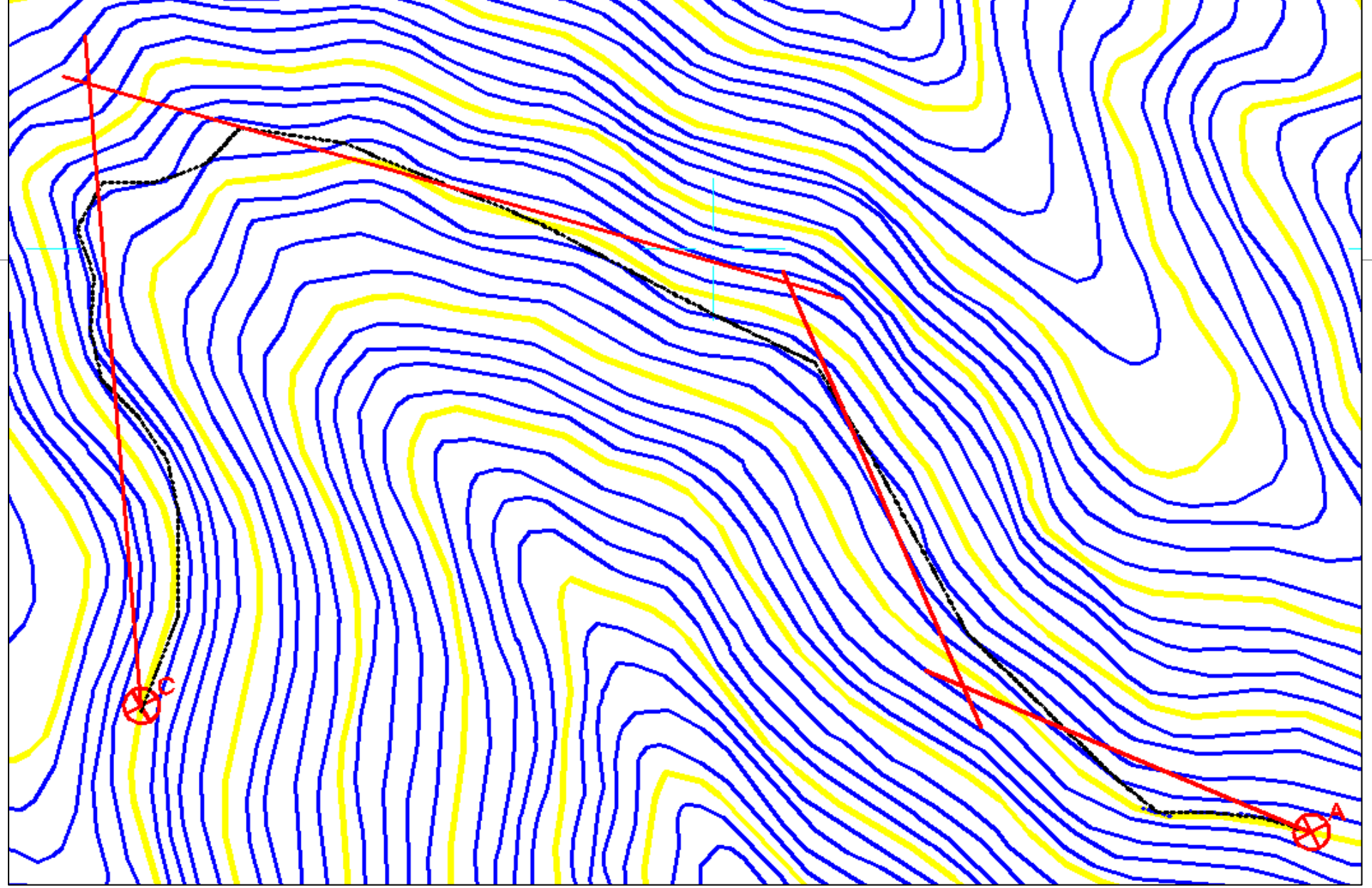


Route Study

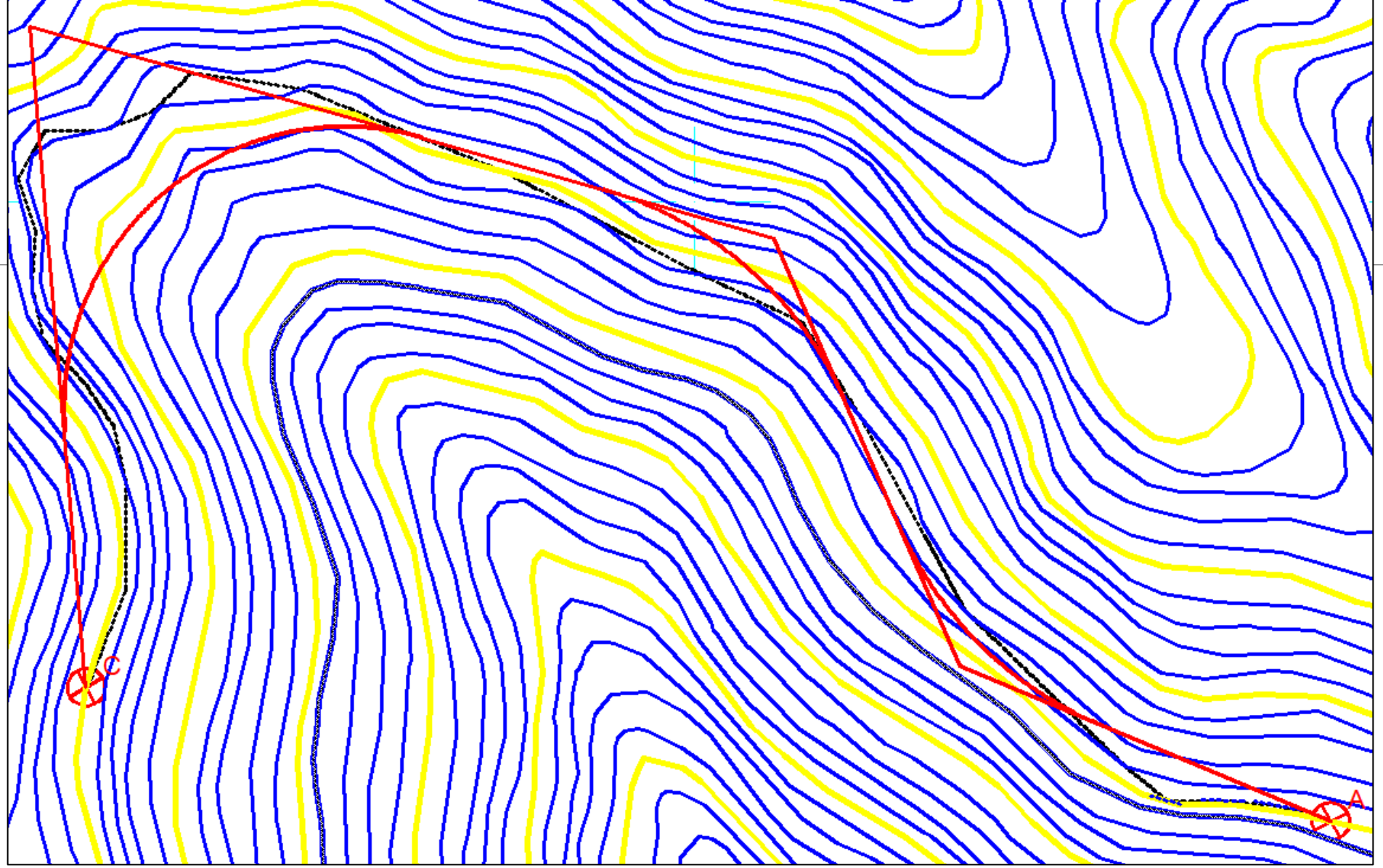
- Line segments (alignments) will be placed on the obtained zero polygon and proper circle arcs (curves) will be placed between the line segments. A distance of 30 m has to be provided between the curves on the same side while 60 m has to be provided between inverse curves.

- Do not place horizontal curve closer than 40 m to the beginning and end points of the highway.

- The radius of the horizontal curves can be 100 m at least in order to provide the standards. Yet sometimes it may be less than 100 m, so you can use 65-70 m for special cases. Beyond this, you can ask your instructor.

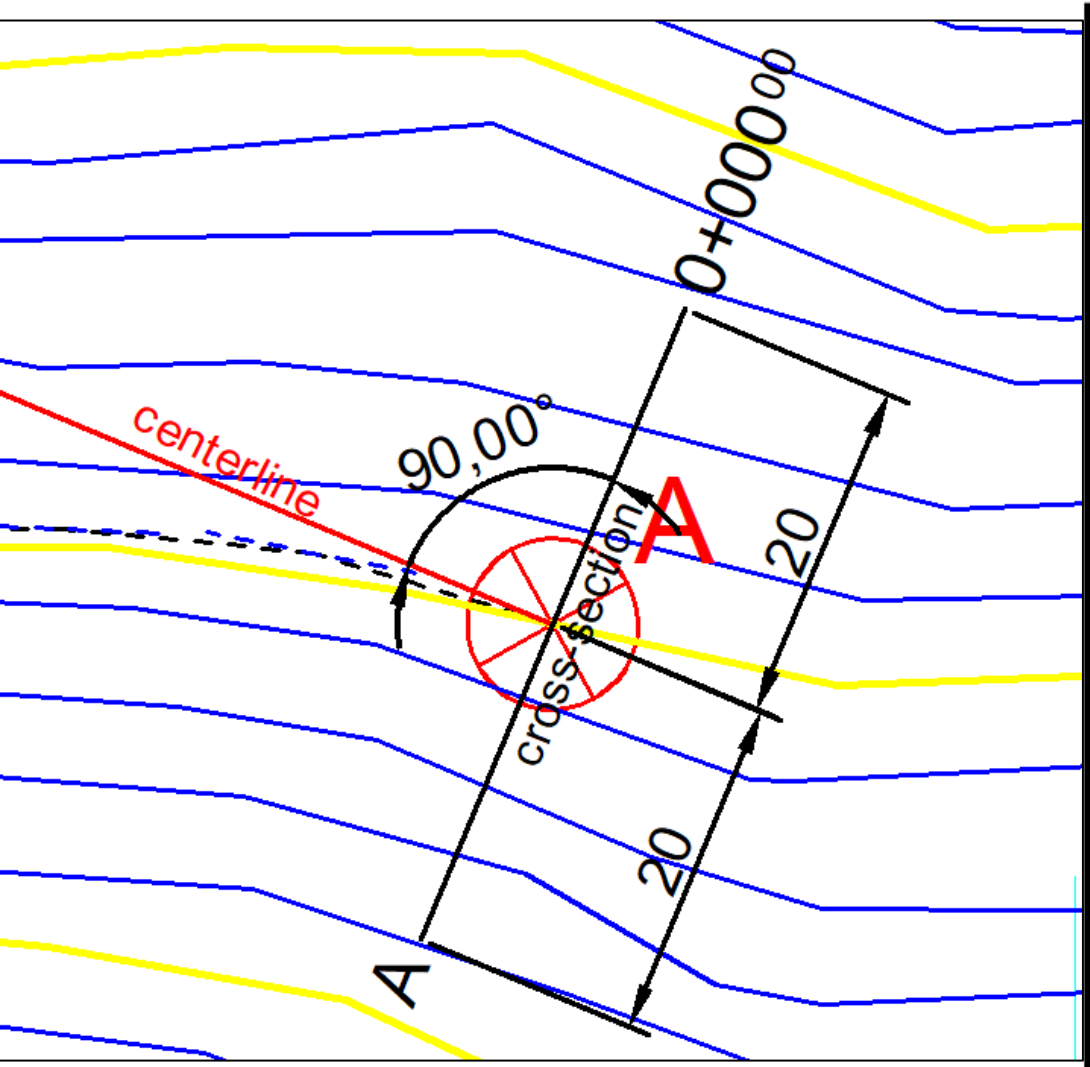


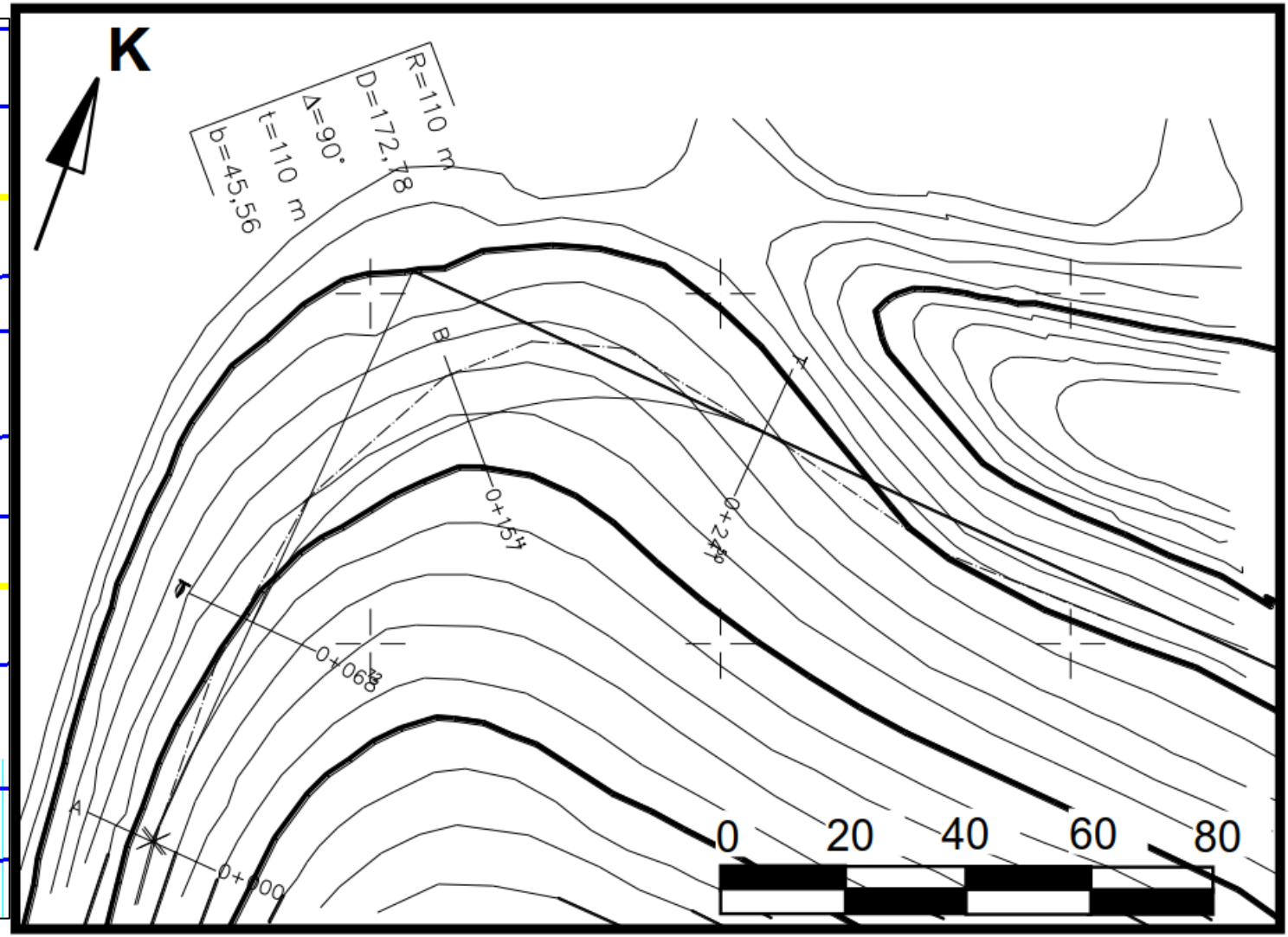
Firstly, while placing the alignments and curves try to draw the near zero line.



Secondly, while placing the alignments and curves try to balance the earthwork as possible as you can do. Which means the areas between the zero line and the route on left and right should be balanced.

- The beginning and ending points of the highway and the beginning, ending and the midpoint of the horizontal curves will be shown on the route.
- On these points, cross-section lines will be drawn vertically to the axis and with the length of the right of way (a total of 40 m, 20 m on right and 20 m on left).
- According to the direction of the highway, the point names will be written on the left side of the cross-section lines while the kilometer values will be written on the right side).
- The kilometer of the start point “A” will be taken as 0+000. The font of the point names and the kilometers will be “Arial” and the font size will be 3 ~ 4 mm.
- The cross-section linewidth will be, 0.25 mm while the zero polygon, alignment and curve linewidth will 0.3 mm.
- The highway platform will not be drawn on this phase. Studies are going to be followed on the route axis.

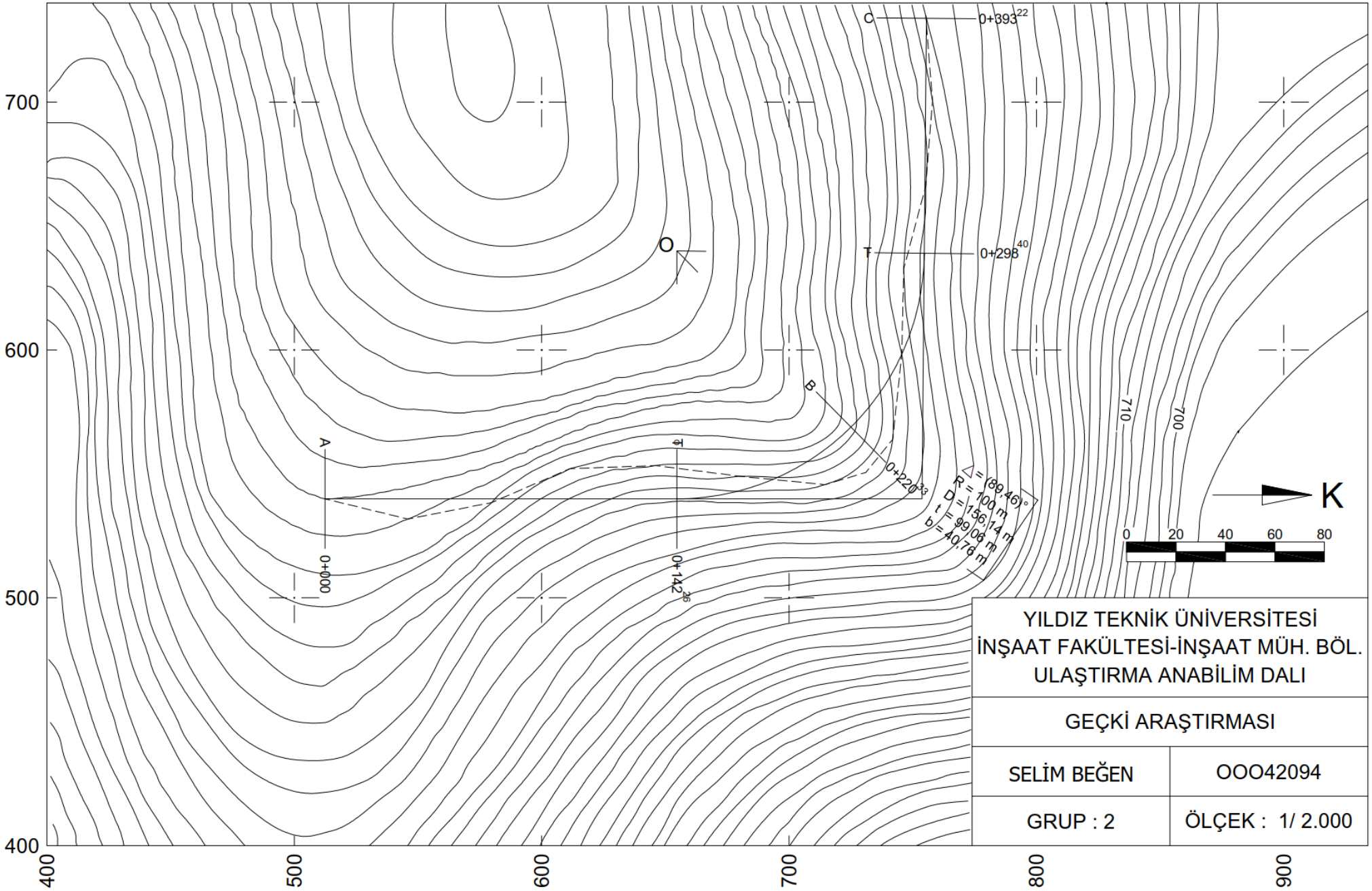




On horizontal curve radius (R), curve angle (Δ), tangent length (t), arc length (D), and bisector values (b) should be on the outside of the curve. Use two digits after decimal for these values.

Cross-sections for route study:

- The beginning (A) and ending (C) points of the highway
- The beginning (Φ), ending (Ŧ) and the midpoint (B) of the horizontal curves



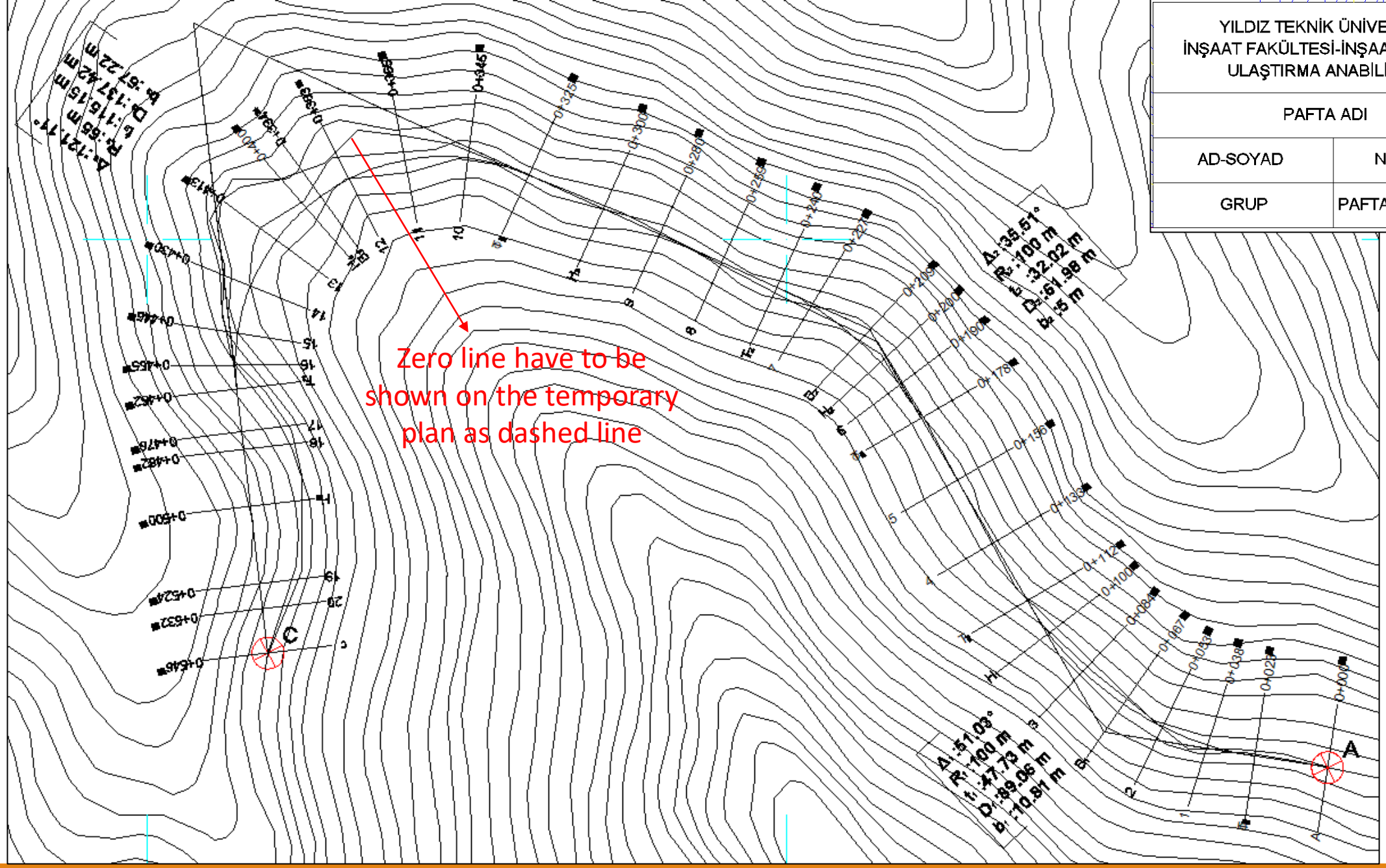
Temporary Plan

Cross-sections will be added for the temporary plan in order of the following:

- The beginning (A) and ending (C) points of the highway
- The beginning (ⱷ), ending (Ⱶ) and the midpoint (B) of the horizontal curves
- Each hectometer (each 100 meters, such as H1, H2, H3, etc.)
- Each point that the route axis intersects with the contour lines
- If the distance between the consecutive cross-sections are longer than 25 m (on an alignment) or 20 m (on a curve), you have to add new cross-sections.

Temporary Plan

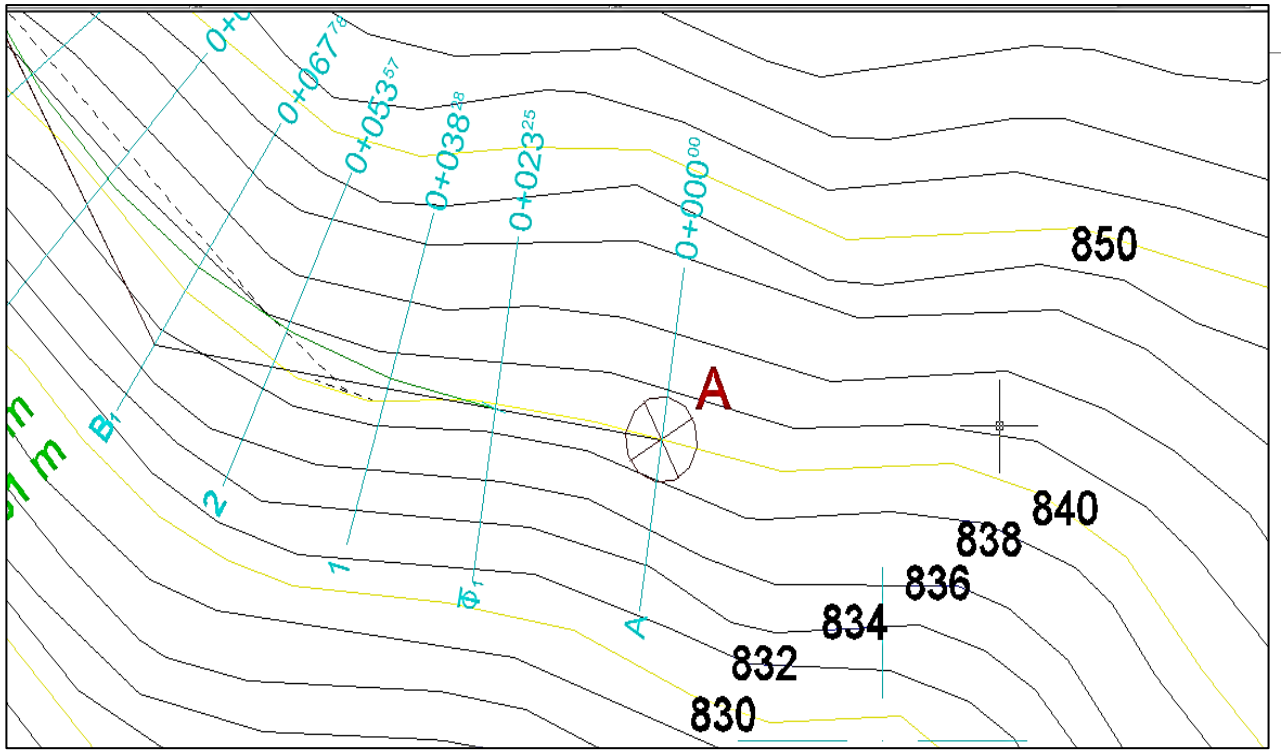
- The cross-sections, except the special points will be enumerated starting from 1, to the end of the highway. All numbers in respect to this will be written on “Arial” font and 3 ~ 4 mm length.
- After all these cross-section lines are drawn, if some cross sections are intersecting with each other or are too close to each other (following each other with a distance less than 5 m), some of them will be cancelled according to their importance.
- The importance is 1. starting and ending points of the highway, 2. cross-sections of horizontal curves, 3. hectometers and 4. others, respectively.

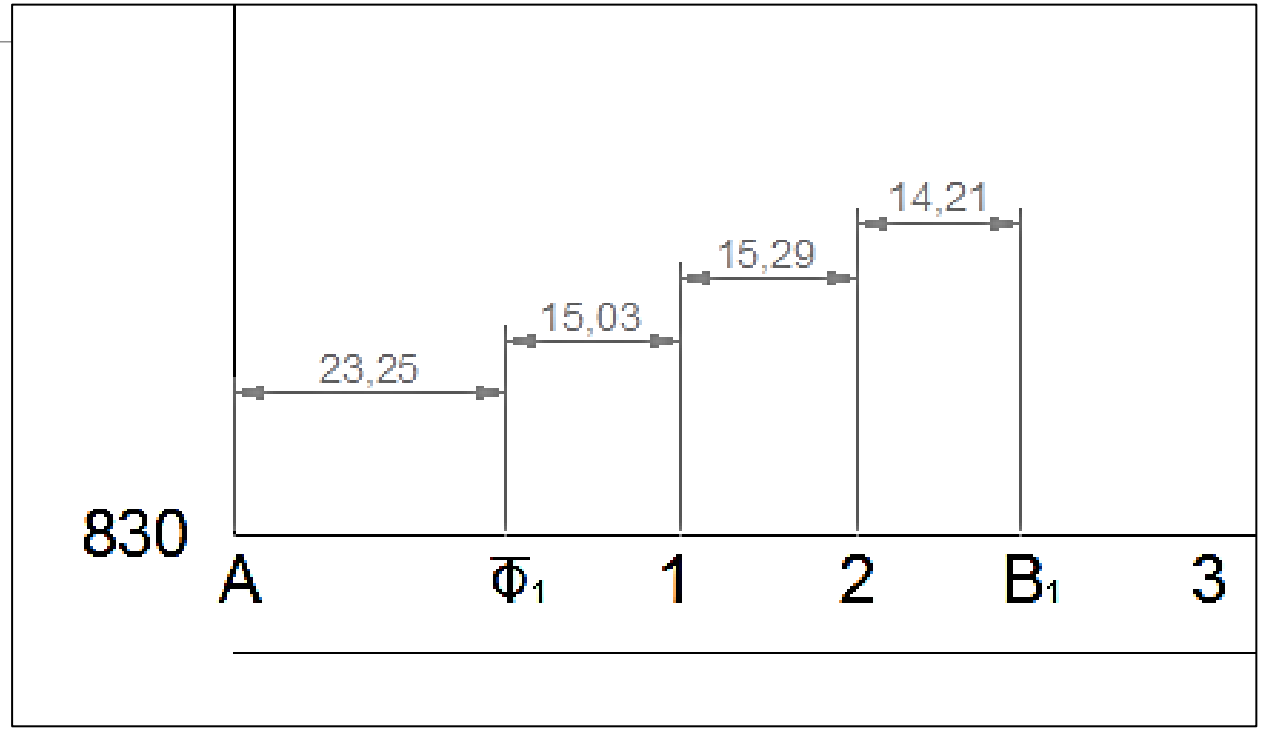


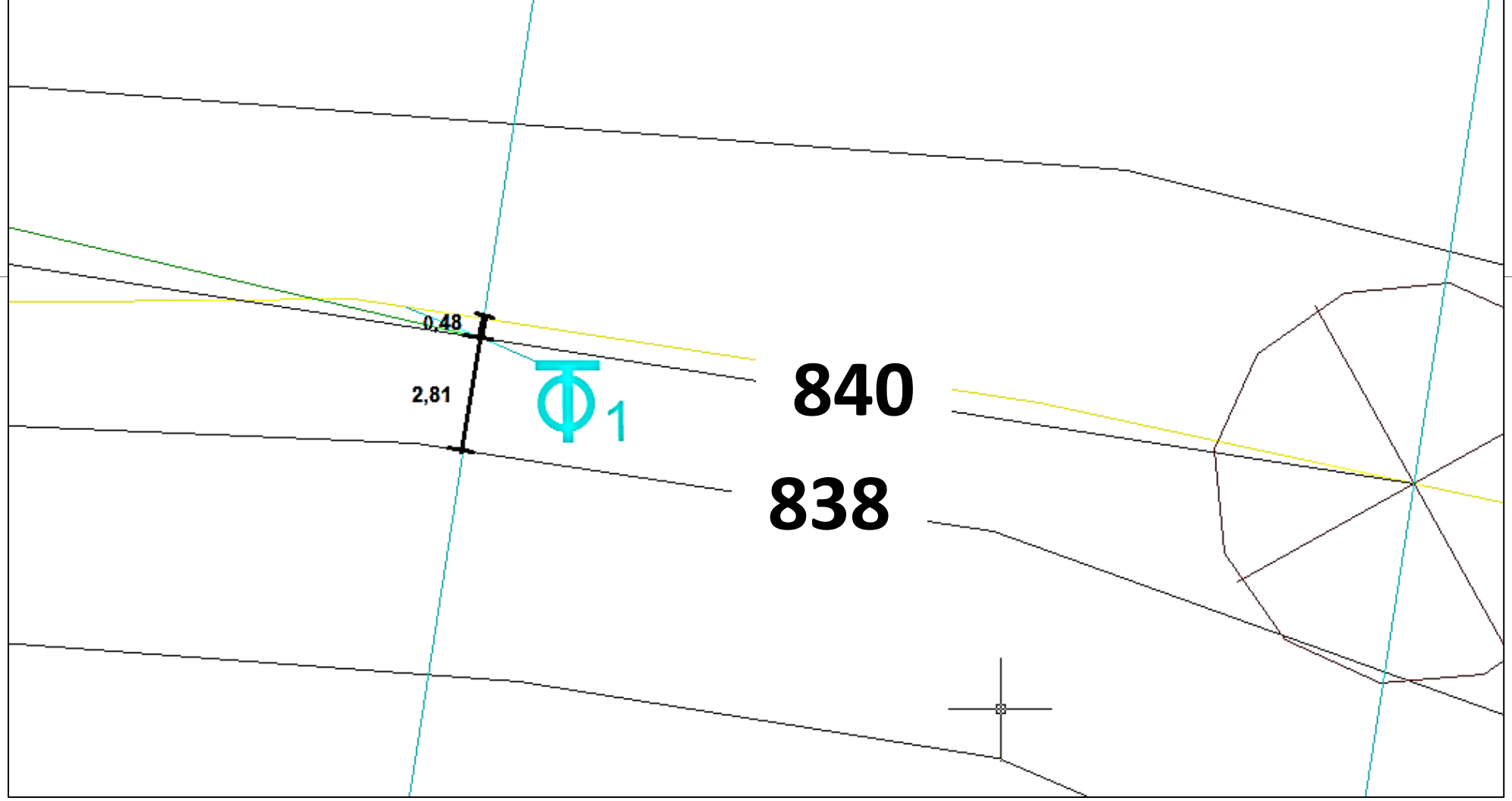
YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PAFTA ADI	$(I_p$=? cm)
AD-SOYAD	NUMARA
GRUP	PAFTA NO:1-1 / 2000

Temporary Profile

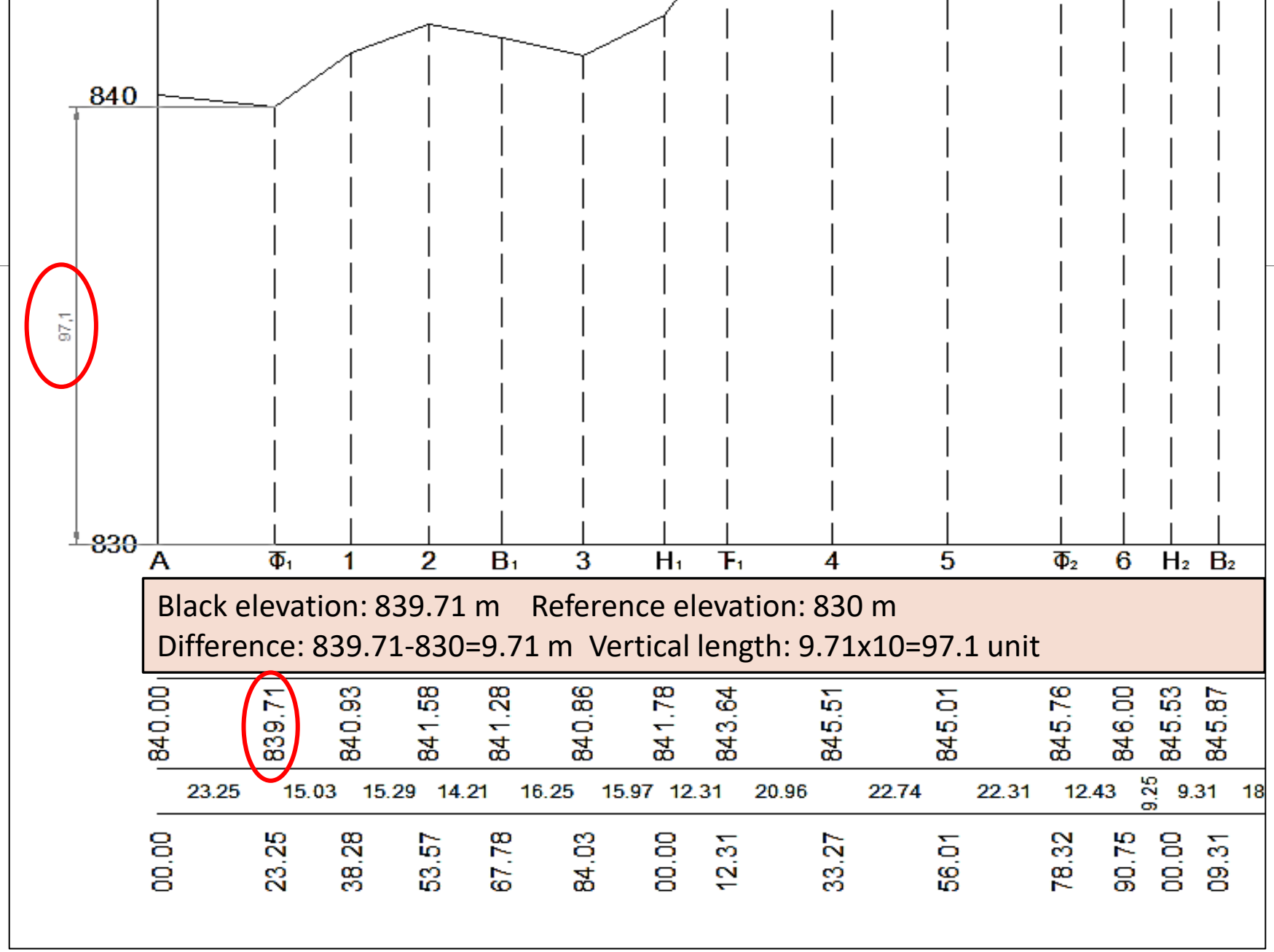
- Profile is the section taken along the axis of the road.
- The horizontal scale of the profile will be taken as 1/2000 and the vertical scale as 1/200.
- Considering the low value of the elevation A or C, the reference elevation for profile will be taken 10 m less.
- To prepare the temporary profile, the black elevations on the center of each cross-section should be known.
- If these points intersect on a contour line, the black line can be taken as the elevation of this contour line, directly.
- For other cross-sections, they have to be calculated using an interpolation. Two decimal values will be taken on the elevation value found on interpolation.


0+067
0+053
0+038
0+023
0+000
850
840
838
836
834
832
830
A
B1
2
1
Ф1
A


14,21
15,29
15,03
23,25
830
A
Ф1
1
2
B1
3



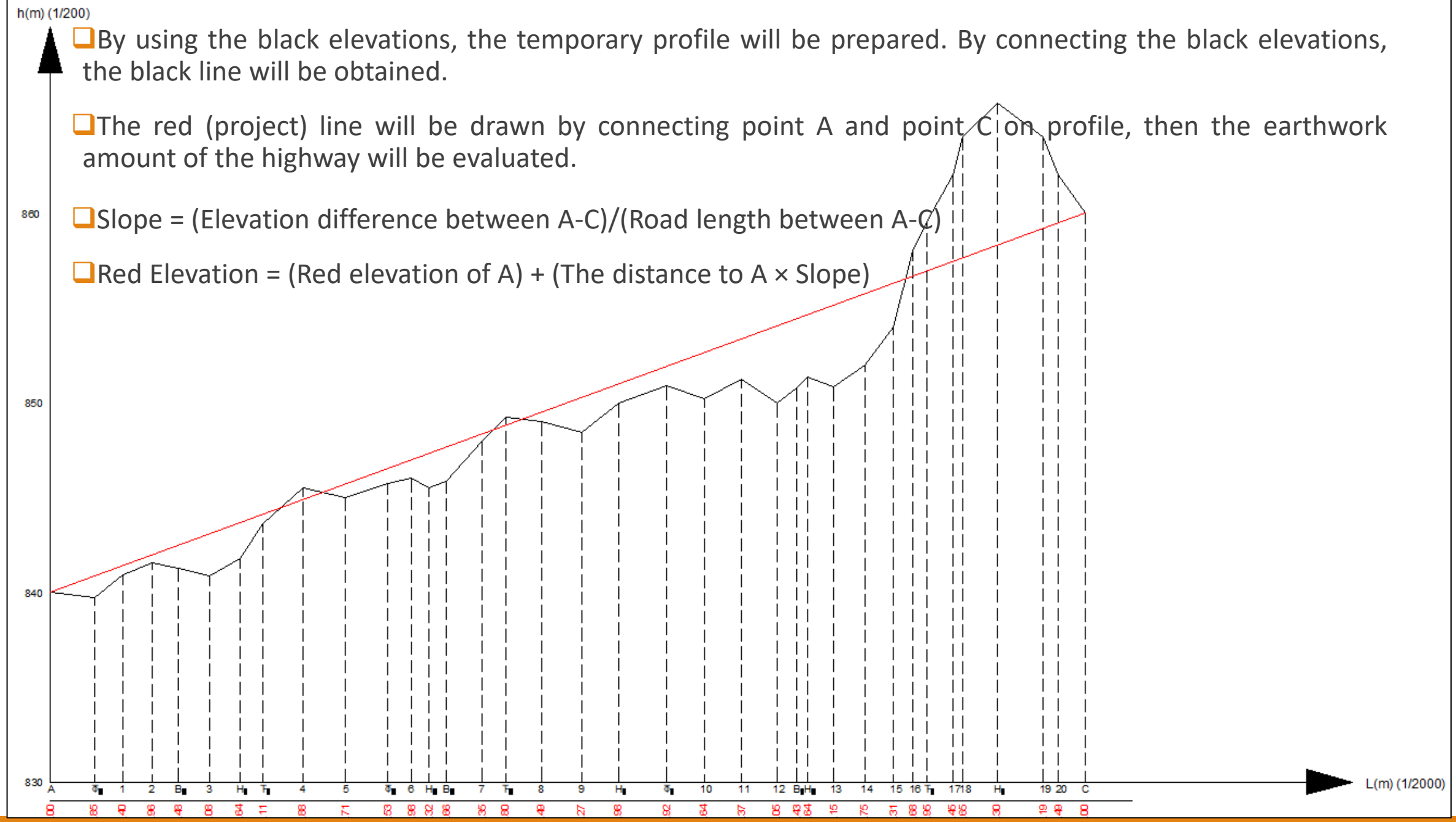
2.81+0.48=3.29 unit length → elevation difference 2 m
0.48 unit length → elevation difference ? → ?=0.29 m
So elevation of the Φ is 840-0.29=839.71 m



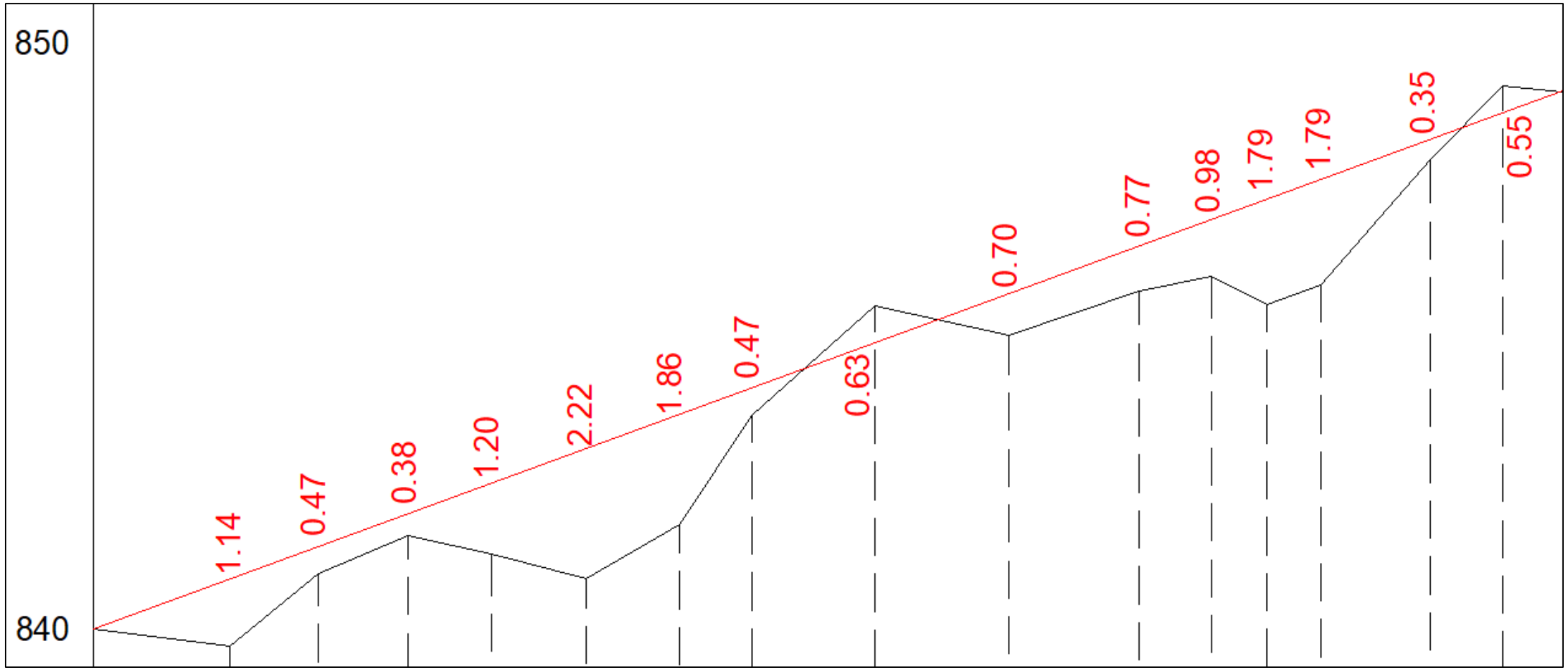
Black elevation: 839.71 m Reference elevation: 830 m

Difference: 839.71-830=9.71 m Vertical length: 9.71x10=97.1 unit

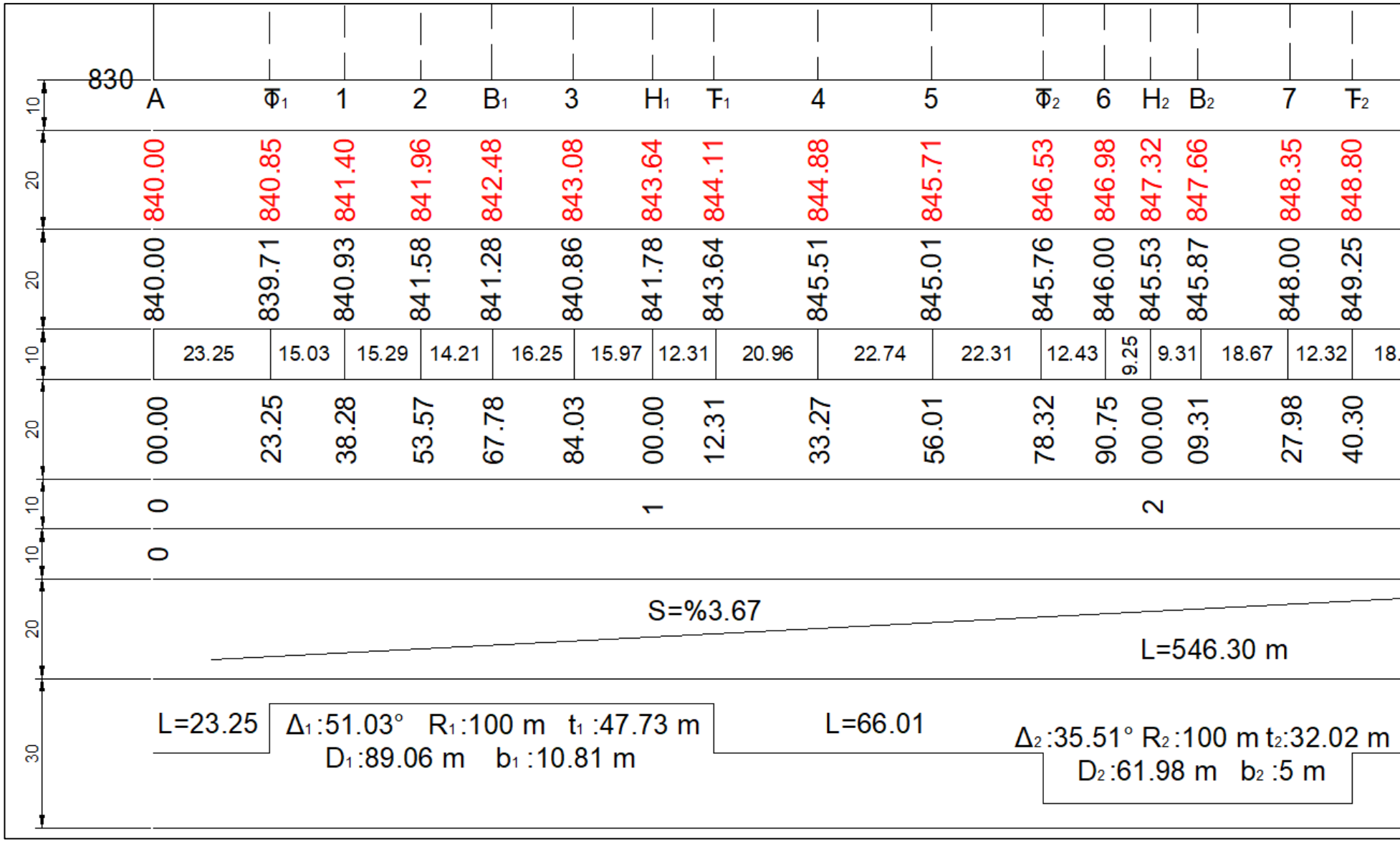
- By using the black elevations, the temporary profile will be prepared. By connecting the black elevations, the black line will be obtained.
- The red (project) line will be drawn by connecting point A and point C on profile, then the earthwork amount of the highway will be evaluated.
- Slope = (Elevation difference between A-C)/(Road length between A-C)
- Red Elevation = (Red elevation of A) + (The distance to A × Slope)



- Also, you should write the elevation differences between red and black line. If it is «fill» it will be written above of the red line; if it is «cut» it will be written bottom of the red line.



- The information which includes “cross-section name, red elevation, black elevation, dıstance between consecutive cross-sections, distance from starting point, hectometer value, kilometer value, grade, schematical plan” will be prepared at the bottom of the profile rows.



First Submission

You have to submit two drawings in one PDF file. The sheets to be sent on the 1st submission:

- Temporary Plan (file that saved as PDF with 1/2000 scale)
- Temporary Profile (horizontal scale: 1/2000; vertical scale: 1/200; file that saved as PDF)
- Zero line study will not be submitted separately. It will be seen on the temporary plan. In order to check if the route study is proper, zero line has to be on the paper.
- Calculations will not be delivered in this submission. You will submit the calculations in the final submission.

Due Date for 1st Submission: 29 November 2021-Monday

Attention to the students who will make 1st submission of the highway assignment!

The assignments will be uploaded to the page https://bit.ly/YTUKarayoluProjesi2021 until **12:00 PM (noon) Monday, 29 November 2021** at the latest. **You must have a gmail account (e-mail) (if you don't have, you have to create an account) in order to upload.**

The sheets to be sent on the 1st submission:

1- **Temporary plan sheet** (file that saved as PDF with 1/2000 scale)

2- **Temporary profile sheet** (horizontal scale: 1/2000; vertical scale: 1/200; file that saved as PDF)

- Save the Autocad drawings of your temporary plan and temporary profile in PDF format and in A3 size with proper scale by using the "Microsoft Print to PDF" tab in the "Plot" section.
- Combine both PDF files (temporary plan and temporary profile) and create a single PDF file.
- Edit the name of this PDF file as “Group5_StudentID_NameSurname” and upload it to the Google form page which given above.

- If you encounter a problem during file upload, you can send your assignment document to karayolu2020@gmail.com until the deadline.
- You can use online PDF combining programs to combine separate PDF files into a single file (e.g. http://combinepdf.com/tr).
- See the assignment information sheet for information about the preparation of these sheets. Especially in the temporary plan, it is sufficient to show the road centerline with a straight line (i.e. the road platform will not be drawn). In addition, the tangents of the curves, points of intersections, the values of the horizontal curve elements (Δ, R, t, D, b) must be shown in this sheet. The zero line must be drawn with a dashed line on the temporary plan and the slope of the zero line and the length of the zero line must be written on the title block.
- The slope of the road must be written in the temporary profile.

Note 1: Assignments submitted after the due date will not be accepted.

Note 2: If you have health problems or a valid excuse, we will request the relevant report in order to give additional time.

Attention to the students who will make 1st submission of the highway assignment!

In the list below, there are possible mistakes that may be done in the preparation of the highway assignment.

- Please be careful to not make mistakes given in this list, while preparing your homework.
- The evaluation of the assignment will be made according to following mistakes. So, please try to do not these mistakes.

Possible Mistakes

Mistake No	Explanation
Mistake No 1	Coordinates were not adjusted according to the student ID.
Mistake No 2	Information in the title block is missing or incorrect.
Mistake No 3	The scale of the temporary plan is incorrect.
Mistake No 4	There was no zero line or it was invisible.
Mistake No 5	Zero line intervals were not equal.
Mistake No 6	Zero line intersected with contour line (or these are tangent) and zero line was not divided to the small part.
Mistake No 7	Zero line was not drawn by dashed line.
Mistake No 8	The slope of the zero line was not written.
Mistake No 9	The route was not close to the zero line and earthwork was not balanced.
Mistake No 10	The distance between the inverse curves was below the minimum value (60 m).
Mistake No 11	The distance between the curves with the same direction was below the minimum value (30 m).
Mistake No 12	Curve radius was selected as less than 100 m, not suitable (if it is not a necessity)
Mistake No 13	Curve radius was not selected as a multiple of 5.
Mistake No 14	The distance between consecutive cross-sections was less than 5m or more than 25m (max 20m in curved section).
Mistake No 15	The beginning or ending points of the curves is coincided with the cross-section A or C.
Mistake No 16	Values of the curve elements were incorrect or missing.
Mistake No 17	Curve elements were calculated but not written on the outside of the curve on the plan.
Mistake No 18	The amount of cross-sections was not sufficient in order to define ground line.
Mistake No 19	Cross-section widths were not as much as right of way.
Mistake No 20	According to the direction of road (from A to C), cross sections names were not written to the left or km values to the right.
Mistake No 21	Cross-sections were not placed perpendicular to the centerline of the road.
Mistake No 22	Platform lines were drawn in temporary plan.
Mistake No 23	The scale of the temporary profile is incorrect.
Mistake No 24	The longitudinal slope was not written on the profile
Mistake No 25	There are missing sections in the table given at the bottom of the profile.
Mistake No 26	Points A and C were not connected by a red line on the profile.
Mistake No 27	The distribution of earthworks was not balanced.
Mistake No 28	The temporary plan sheet was not submitted.
Mistake No 29	The temporary profile sheet was not submitted.
Mistake No 30	Other